



SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

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Write a Java program to demonstrate how to use synchronization to prevent multiple threads from accessing a shared resource simultaneously.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2
3     // Shared resource that should not be accessed by multiple threads at the same time
4     static class SharedResource {
5         // Synchronizing on this object to ensure only one thread enters at a time
6         public synchronized void access(String threadName) {
7             System.out.println(threadName + " is trying to access the shared resource...");
8             try {
9                 // Simulate some work with the shared resource
10                Thread.sleep(1000);
11            } catch (InterruptedException e) {
12                Thread.currentThread().interrupt();
13            }
14            System.out.println(threadName + " has accessed the shared resource.");
15        }
16    }
17
18    // Worker thread that tries to access the shared resource
19    static class Worker extends Thread {
20        private final SharedResource resource:
21
```

OUTPUT

```
Thread-1 is trying to access the shared resource...
Thread-1 has accessed the shared resource.
Thread-3 is trying to access the shared resource...
Thread-3 has accessed the shared resource.
Thread-2 is trying to access the shared resource...
Thread-2 has accessed the shared resource.
All threads finished execution.
```

**INPUT**

Your INPUT goes here!

